Setting up the HMG Python Module

The supplied “hmg.zip” on Moodle contains the pure Python implementation of the HBV001a rainfall-runoff model. The model is needed throughout this course, so it is necessary to get it running.

Follow the steps below for the setup of the hmg module inside your Python installation:

1. Extract the contents of hmg.zip to a location that is on PYTHONPATH. Ideally, it should be extracted to the site-packages directory of your Python installation. The site-packages directory is located inside the “Lib” directory of your Python installation. Make sure that hmg is not extracted to a new hmg directory i.e., its path should be site-packages/hmg and not site-packages/hmg/hmg.
2. Start a Python interpreter and type “import hmg” and press enter. If the hmg module has been extracted and put in the required location correctly then the interpreter should not show an ImportError. In case it does then speak to me.

Now that the model import works, we need to be able to run it. To do so follow the steps below:

1. Navigate to site-packages/hmg/test on your PC.
2. Copy the test_model.py Python script to a proper (and different) location such as directory of an assignment of this course. Don’t make any other changes to aa_run_model.py inside the hmg module. It will be needed later in case the copy of the original is needed.
3. Rename the copied file to a more appropriate name e.g., aa_run_hbv.py. This will help with remembering what the script is supposed to do.
4. Open aa_run_hbv.py in your desired IDE such as Visual Studio or Eclipse or Spyder by creating a new project for it.
5. Change the value of the variable “main_dir” in the main function to a location with the input data for the model. The input data is provided on Moodle.
6. Now, run the script.
7. If everything is done correctly, a figure of two hydrographs should show up. Close it. Now another figure with all the internal variables of HBV001a should pop up. Close it.
8. At this point the script should stop running. Everything went as planned.
9. If an error shows up then please speak to me.

Try to read the aa_run_hbv.py script and understand what all the variables mean. Discuss with me if there are any confusions or questions.